

Abstract

FitGenie AI is an intelligent, agent-based fitness assistant designed to provide users with personalized and adaptive fitness guidance through natural language interaction. Unlike traditional fitness applications that offer static workout plans, the proposed system utilizes a **Large Language Model (LLM)** in combination with **AI agents** to deliver dynamic workout routines, nutrition recommendations, progress monitoring, and fitness coaching tailored to each user's goals, fitness level, available time, dietary preferences, and physical limitations.

The system employs **LangChain** as the orchestration framework to coordinate multiple specialized AI agents, each responsible for a specific task such as workout planning, nutrition guidance, progress evaluation, and fitness-related question answering. By integrating **Retrieval-Augmented Generation (RAG)** with a trusted fitness knowledge base, the agents retrieve relevant information to generate accurate, context-aware, and personalized responses while minimizing misinformation. User profiles, conversation history, and fitness progress are maintained to enable the agents to continuously adapt recommendations as user requirements change over time.

Through collaborative AI agents, the application functions as a virtual fitness coach capable of updating fitness plans, answering user queries, tracking achievements, and providing motivational support throughout the fitness journey. The proposed system combines Generative AI, multi-agent architecture, conversational intelligence, and intelligent information retrieval to create a scalable and user-centric fitness platform. By delivering adaptive and personalized guidance, FitGenie AI aims to make professional fitness coaching more accessible, interactive, and effective for users with diverse lifestyles and fitness goals.